3D Items Part 4a – Clothes Topics

Wogrim's Epic Guide to Clothes Topics with Blender and KK Modding Tools

Before Reading This Guide

Read the previous guides. Clothes have a lot of special considerations; you want to be as comfortable as possible with making 3D items in order to minimize the difficulty of Clothes.

What Are We Making?

The main difference between Clothes and other item types is that Clothes are skinned to the character's skeleton instead of their own skeleton. It is easy for this to cause clipping problems when someone changes sliders in Maker, or from animations, which can make Clothes very difficult to get right.

But before we get to making Clothes, there are a lot of things to consider. Some apply to all Clothes, and some apply only to certain types of Clothes:
- the clothes MB
- clothes color pickers and shaders
- half states
- alpha mask
- emblems
- optional parts
- decorations
- liquids (semen)
- dual slot items
- skin-tight clothes
- _low version
- adding extra bones
- accessory clothes

Then we'll look at the general process for making Clothes and do a quick bra mod.

The next guide will be more focused on the making of the clothes, but I promise this guide is full of information you should know first.

The Clothes MB

The Clothes MB is called **ChaClothesComponent** and is an important part of getting many of the Clothes features to work. But the thing is, with some DLC the game devs added a new version of the Clothes MB to the game, which works slightly differently and has extra features (something with sleeves, not sure what else). I use KK Party + After Party DLC (Steam version of the game) + HF Patch; as far as I can tell the extra features don't work for me.

So I can't confirm how those new features work, but more importantly, I couldn't get optional parts to work on the new MB. Why was I using the new MB? Because (currently) KK Modding Tools uses the new Clothes MB. So if you find the MB in KK Modding Tools is not working for you, your options are:

- after you make your mod, use SB3UGS to copy over the old Clothes MB from a game item and use that instead (and don't forget to compress with LZ4 high)
- edit the ChaClothesComponent script in KK Modding Tools to the old Clothes MB (experimental)

Also, even if the optional parts work for you on the new Clothes MB, you might still want to use the old Clothes MB so that they work for everyone.

Clothes Color Pickers and Shaders

For the color pickers, Clothes work a little different from other items in that the MainTex and ColorMask are specified in the list file. The game mixes the MainTex and the color picker colors (including patterns) and gives that to the shader. Because of this, there is no slot for the ColorMask on the game's clothes shaders, but that also means the color pickers will work for any shader that uses MainTex. However, it is still generally recommended to use the game's clothes shaders because of other features they have.

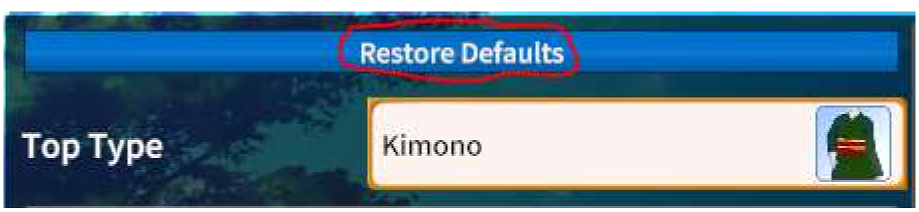
The game's shaders for clothes are:

- Shader Forge/main_opaque - use for solid clothes; transparency on MainTex cuts out parts of the cloth
- Shader Forge/main_opaque2 - same as above, but doesn't draw back faces of the mesh; not used much
- Shader Forge/main_alpha - used for semi-transparent clothes; partial transparency on MainTex makes see-through clothes; one-sided so you have to duplicate & flip mesh faces if you want to see both sides; unable to receive shadows; increased performance cost so don't use for solid clothes

If you have a more complicated item that needs a second MainTex and ColorMask, there are MainTex02 and ColorMask02 slots in the list file; the mixed texture gets sent to the meshes in **rendNormal02** in the Clothes MB instead of **rendNormal01** (where you normally put all your meshes).

When filling out the list file, you must fill in both MainTex and ColorMask, or neither; it won't work properly if you just fill in one of them. So if you want your item to not be colorable, make a black ColorMask and put MainTex and ColorMask in the list file, OR put neither in the list file and put the MainTex on the material. If your item should be colorable, be sure to make a big white MainTex of resolution that matches the ColorMask. Also, be sure to turn on **useColorN01**, etc. for each color picker your item needs to show.

At first glance (2 years for me), clothes do not appear to have default colors because the color picker colors stay the same when switching between items. However, the "Restore Defaults" button above where you select a clothes item will set the colors to that item's default colors specified in the MB (**defMainColor01**, etc.).



Half States

Clothes half states are the partially-undressed versions of the Clothes item. These are separate meshes which are shown and hidden by the game; the game finds them by looking for certain transform names in the item's hierarchy, similar to how it looks for N_move and N_move2 for Accessories. The Clothes slots that allow half states are as follows, and need meshes under appropriate transform names to show/hide properly:

- underwear and bot
    - **n_bot_a** - full state
    - **n_bot_b** - shift state
    - **n_bot_c** - hang state (underwear only)
- stockings
    - **n_panst_a** - full state
    - **n_panst_b** - shift state
- bra and top
    - **n_top_a** - full state
    - **n_top_b** - shift state
- school uniform pieces
    - same as top? (not fully tested)

If you have meshes that are not under these transform names, they will be shown in all states (except off). Also, if your item doesn't have half states, you can tell the game with the "State Type" column in the list file.

Half states usually use the same material (just a different mesh, which still goes in rendNormal01), but you can have them use the second set of textures (and put them in rendNormal02). You might do this if you want to do something different with the textures, such as a ripped clothing look.

<u>Alpha Mask</u>

AlphaMask is a texture slot on skin and clothes shaders, which is used to hide parts based on the layer above (specified in that item's list file); this is used by the game to help prevent clipping (and avoid weight painting). So, certain clothes items need to be UV'd similar to the game's items so that the wrong parts aren't shown/hidden at the wrong times:

- bra - so that it can receive an AlphaMask called OverBraMask from a top
- inner uniform top - so that it can receive an AlphaMask called OverInnerMask from an outer uniform top

The game looks for GameObjects with certain names to give them the appropriate AlphaMask. The bra's must be named **o_bra_a** or **o_bra_b**; anything else (as far as I know) will not receive the AlphaMask, even if it uses the same material. The game items use these names for full state and half state meshes; I recommend you do the same. I haven't tested inner tops, but they probably have to be **o_inner_a** and **o_inner_b** based on game items.

In addition to the bra and inner masks, some items can specify an AlphaMask called OverBodyMask which will hide parts of the body. If you need to specify masks in your item's list file, you can point to the game's masks (which are all in ABs starting with "mt_mask") or create your own.

Here's the bra mask that the kimono (top) sets to the bra AlphaMask:

The kimono hides almost everything in full state, and shows the boobs / upper chest in shift state.

The colors used in an AlphaMask are:

- yellow - always show this part of the item
- green - hide this part of the item if layer above is in full state, show if layer above is in half state (or off)
- black - always hide this part of the item (unless layer above is off)

If you need to create your own masks, you can use lower resolution than other textures.

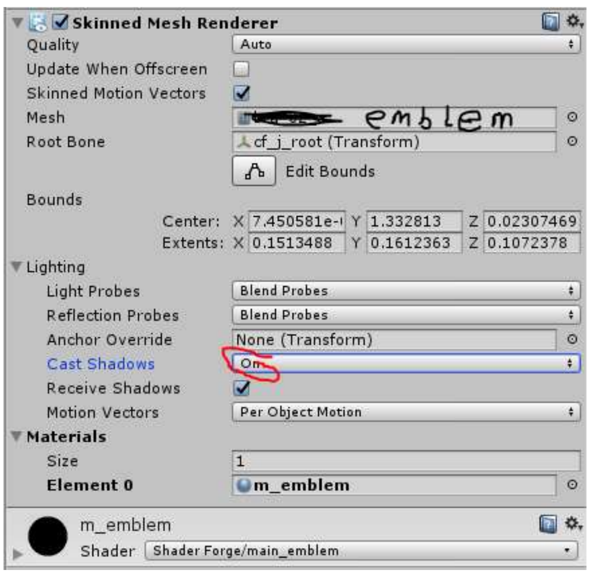
Emblems

Emblems are those images you can put like a school logo on the school uniform outer jackets or the emblem "bra". They go on a square-ish mesh that you place on top of the clothes (if you want it to fit well, just duplicate appropriate faces of the clothes mesh and move it out very slightly). The mesh still gets parented to the appropriate transform name for half states (if you're doing those) but you are limited to two emblem meshes, which go in **rendEmblem01** and **rendEmblem02** in the MB; you should use these for full state and shift state. If you want the emblem in multiple locations, you just have more parts of the same emblem mesh in each location and they all UV to the whole texture.

I know of 2 shaders used for emblems:

- Shader Forge/main_emblem - emblem is given to the MainTex
- Shader Forge/main_emblem_clothes - same but also accepts an AlphaMask
    - note: on the emblem "bra" the alphamaskuv1 variable is set to 1 which moves/scales the AlphaMask (which is the same as what a regular bra receives) to roughly the right size and location to hide the emblem if the nipples are supposed to be hidden; I don't think it works properly if the emblem is not on the nipples

You might also turn off shadow casting for the emblem so people don't notice it is floating above the cloth.



Optional Parts

Clothes can have optional parts that a player can toggle off in Maker. It's a little more work for you, but it's a very convenient customization option to allow people to only wear part of your item just by unchecking a box.

You tell the game your item has optional parts by setting **useOpt01** (and/or **useOpt02**) in the MB to 1. The newer Clothes MB does not have useOpt01 or useOpt02; I assume this is why I couldn't get optional meshes to work with it. The optional meshes must be put in **objOpt01** or **objOpt02** in the MB for the game to show/hide them with the checkboxes in Maker, but they must also go in rendNormal01 or rendNormal02 for color pickers / texture stuff, and they must be placed appropriately in the hierarchy for half states.

Decorations

The mesh for "decorations" in the school uniforms is put in **rendAccessory** instead of rendNormal in the MB, and get the extra "Accessory" color picker color, similar to built-in hair accessories. But for clothes, they can have a pattern, and they use a Clothes shader.

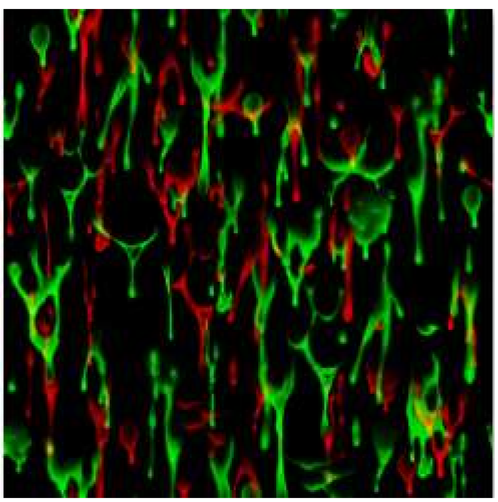
As far as I know, rendAccessory doesn't do anything in other Clothes slots.

Liquids (Semen)

Some sexual actions involve squirting semen on a girl; the game tells the skin and clothes shaders when this happens and the shaders draw liquids in the appropriate places. For making clothes, this means you have 3 textures to worry about:

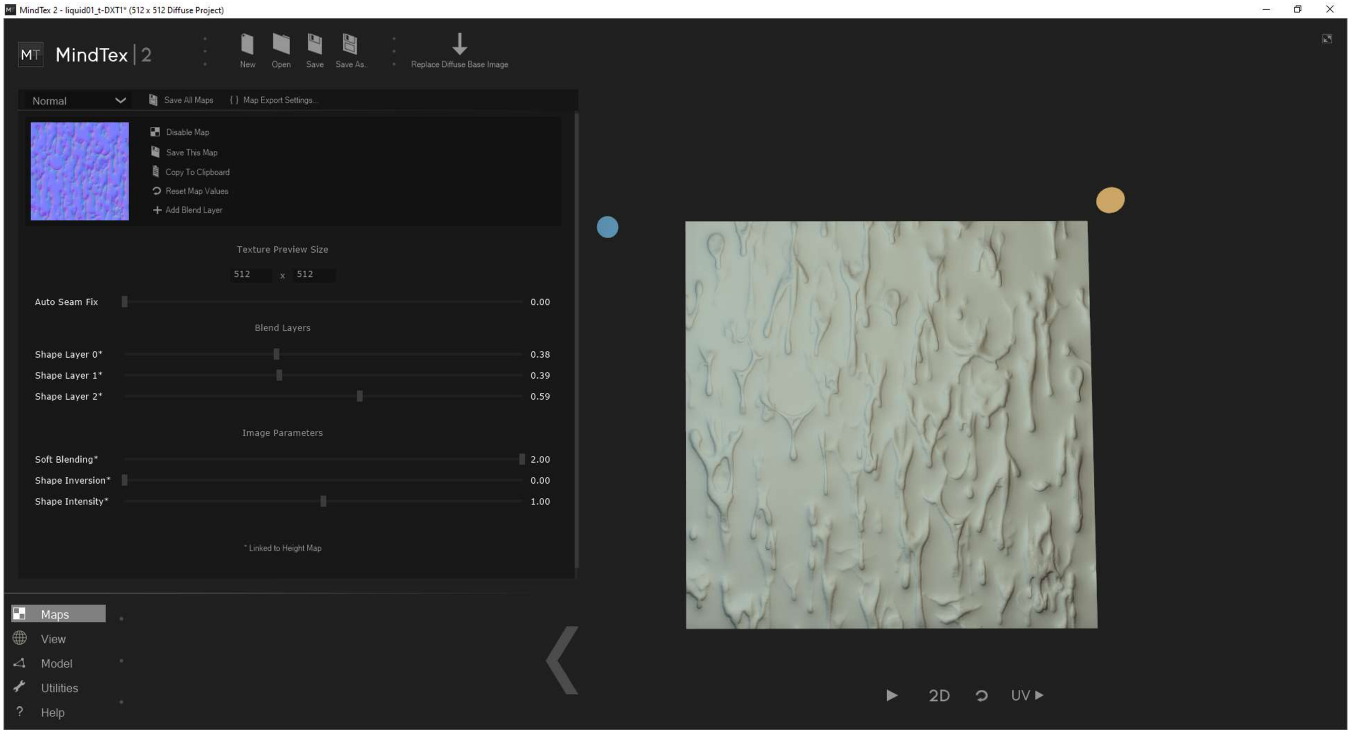
- liquidmask - where on the item the liquid spatter pattern is shown, colored for regions for different sex acts
    - black - no liquids
    - red - front top
    - blue - back top
    - green - front bottom
    - yellow - back bottom
    - teal - face (used by skin; haven't tested if this works on clothes shader)
- Texture2 - liquid spatter pattern
    - black - no liquids
    - red - first layer of liquids
    - green - second layer of liquids
- Texture3 - liquid normals

You can make a custom liquidmask very easily, but making a custom liquid spatter pattern is more involved. I recommend you try to use the generic liquid spatter pattern and normals, which are used by 99.9% of game items (tiled to appropriate size). The spatter pattern looks like this:



To make the generic liquids work with your item, you usually have to tile them so that they're smaller, and your item's UVs (for all the areas that will show liquids) need to be properly rotated, relatively unstretched, and not broken into a bunch of separate pieces. So it may not be an appealing choice if you have to re-UV your item.

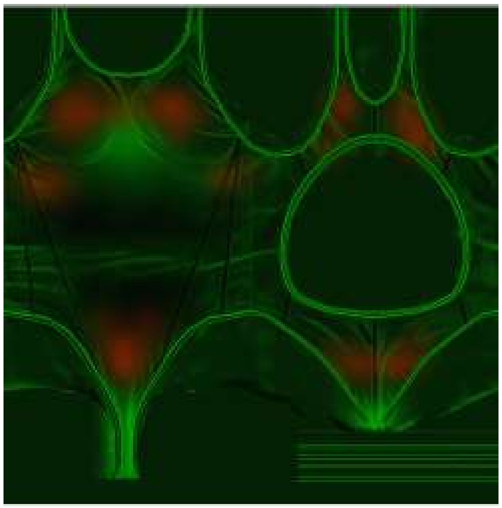
If you decide to make a custom liquid spatter pattern, you *could* make it in 3D and bake proper liquid normals, but it is easier to just draw the liquid spatter pattern and then use software that generates a normal map from it. Here's liquid normals for the generic liquids, generated by MindTex 2 after playing with the sliders (to make it look more liquid-ey).



There's other software (some are free) that will do the same thing (search for "normal map generator"), but I haven't tried them. Be careful, there's different normal map formats; you want the blue-ish type, green = up. But don't stress too much about the liquid normals, it's hard to tell when they're messed up.

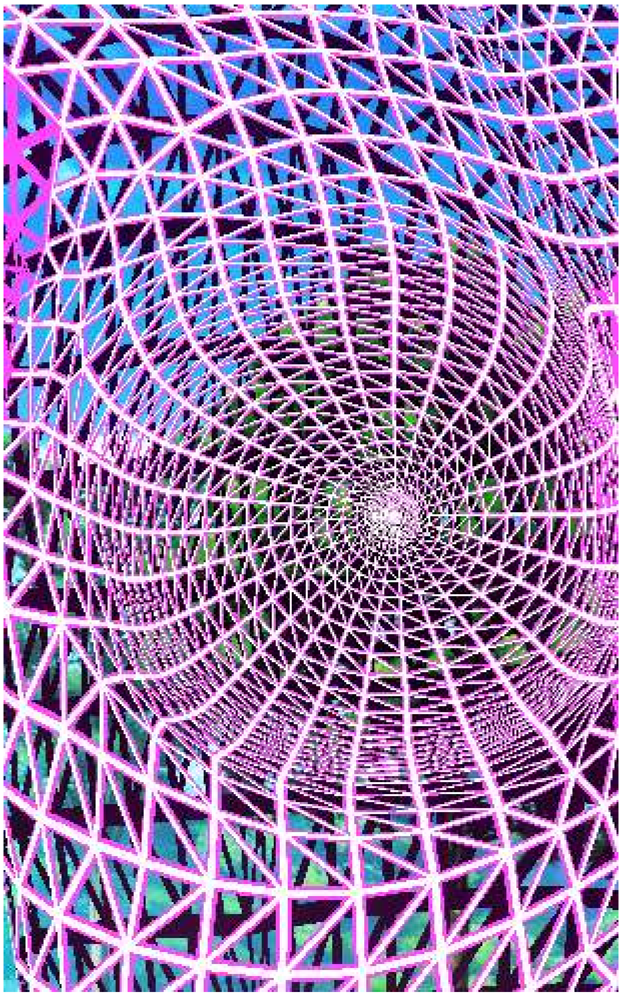
Dual Slot Items

So that the game can have items like one-piece swimsuits and dresses, there are ways to make items that are bra + underwear or top + bot slots. You should study such items and their list entries if you wish to make them. Usually the upper and lower part are separate meshes so that they can separately be put in half state. There's a special way the bra mask can be applied to the top half of the texture space if you have the bra and underwear part in the same texture.



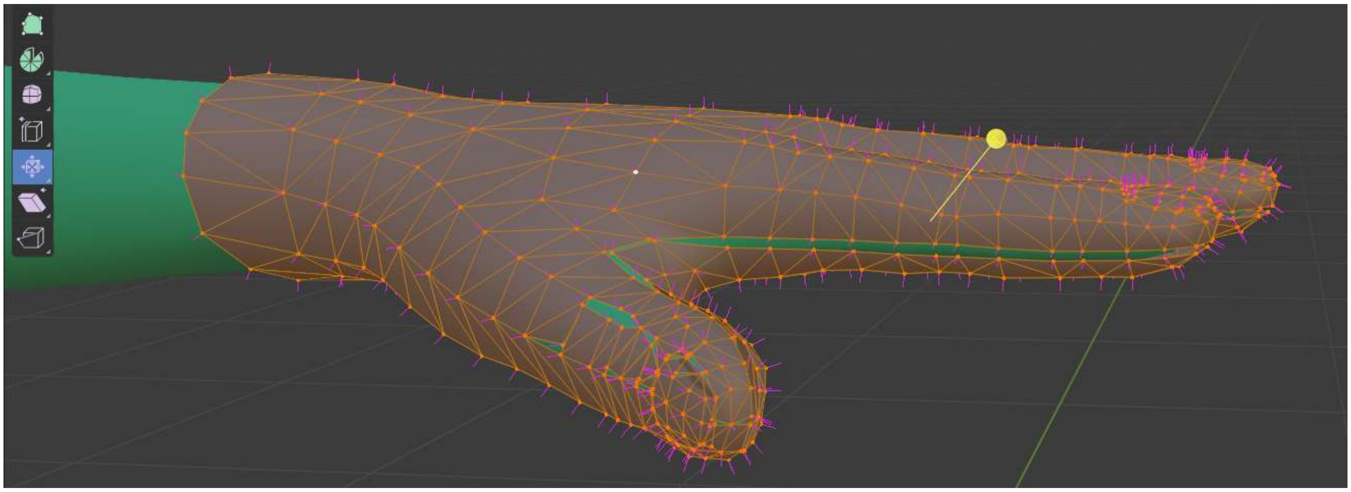
Skin-Tight Clothes

It is generally easy for skin-tight clothes to clip through the skin (from animation and/or Maker sliders), just because they are so close to the skin. In some situations you can avoid the problem with a body mask, but even so you really want the clothes to move how the skin does anyways. Depending on exactly how tight the item will be, you may be able to do this with bone weights alone, but the recommended method is to have the mesh structure of the Clothes item match the mesh structure of the body (just be slightly above the skin). Here's part of the game bra mesh (white, thinner wireframe) with the body mesh (pink):

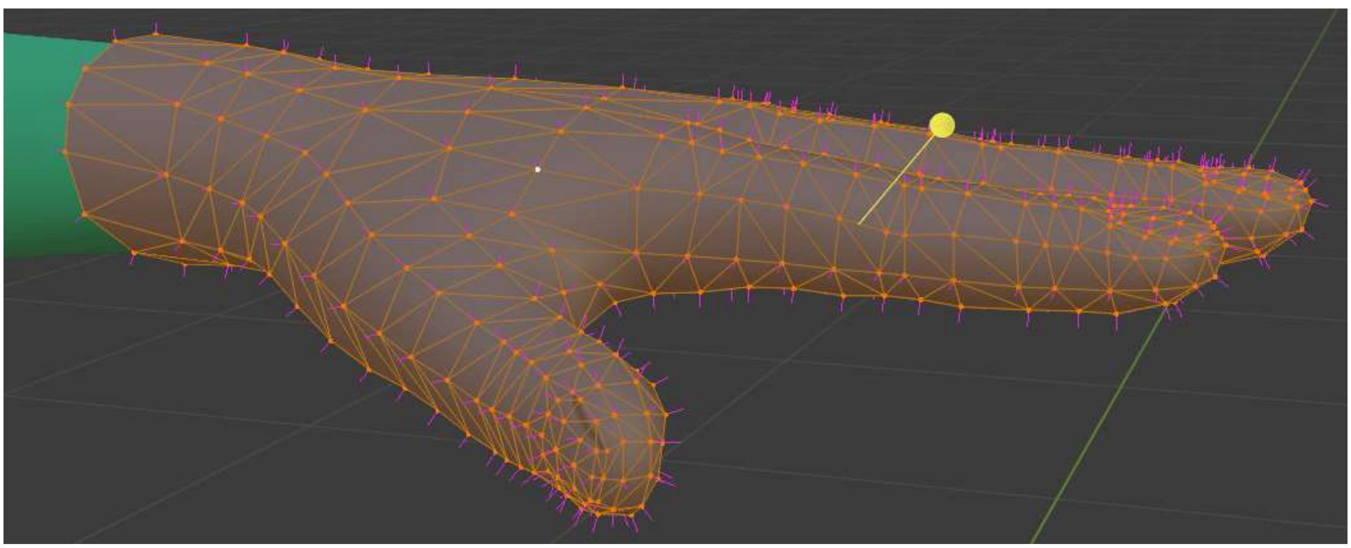


Because the bra mesh lines up perfectly with the body mesh and uses the same bone weights, it is almost impossible for it to clip with the boobs (otherwise it would happen all the time with boob sliders and boob physics). Are all those polygons really necessary? Yes. Otherwise giant boobs will give you problems, and you also won't have a good smooth boob outline/silhouette.

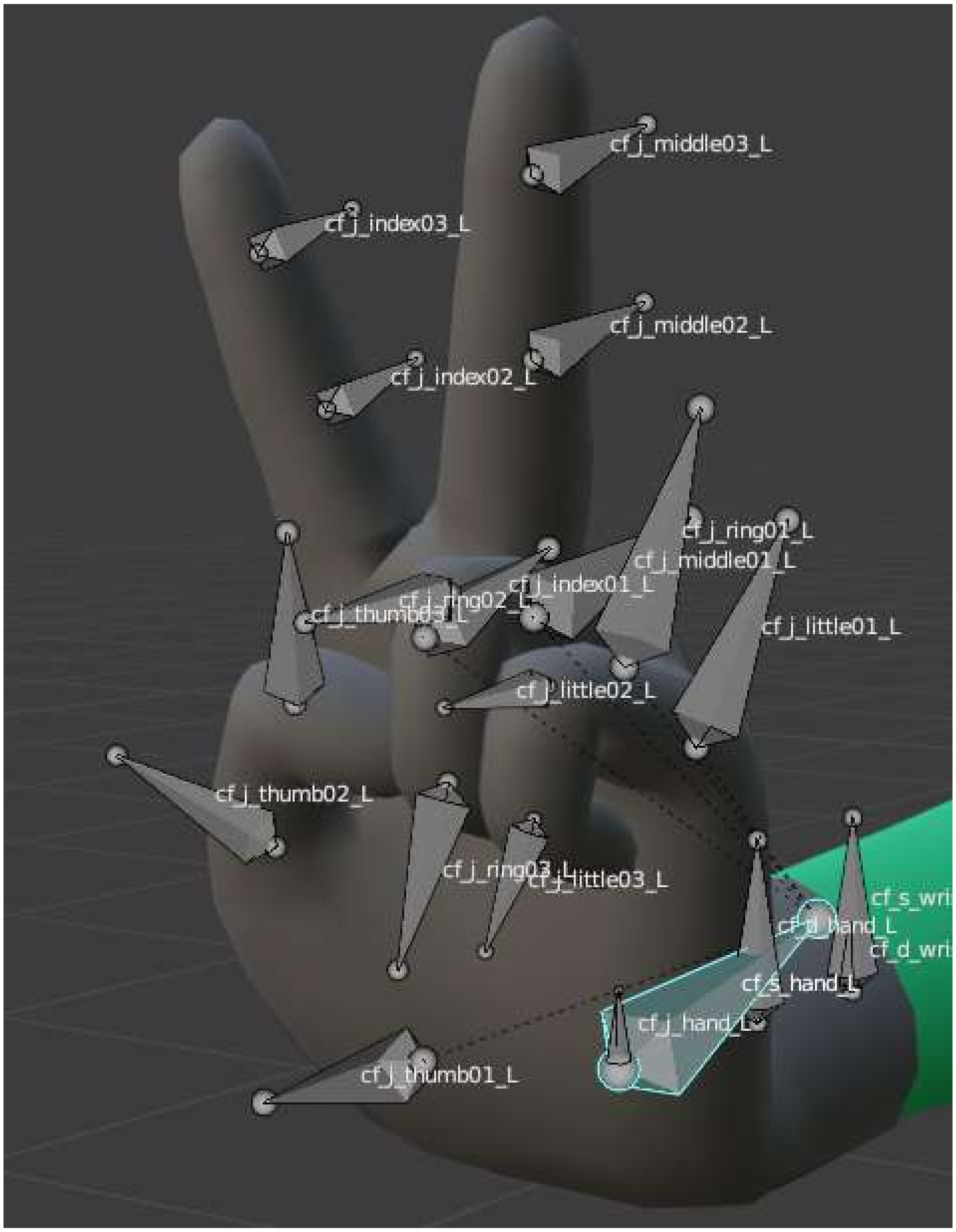
It might seem like you could do this with scaling a copy of that part of the body mesh, but Blender has a better tool for this called Shrink/Fatten, which seems to basically move vertices along their normals. You may notice the mesh coming apart on you when you do this:



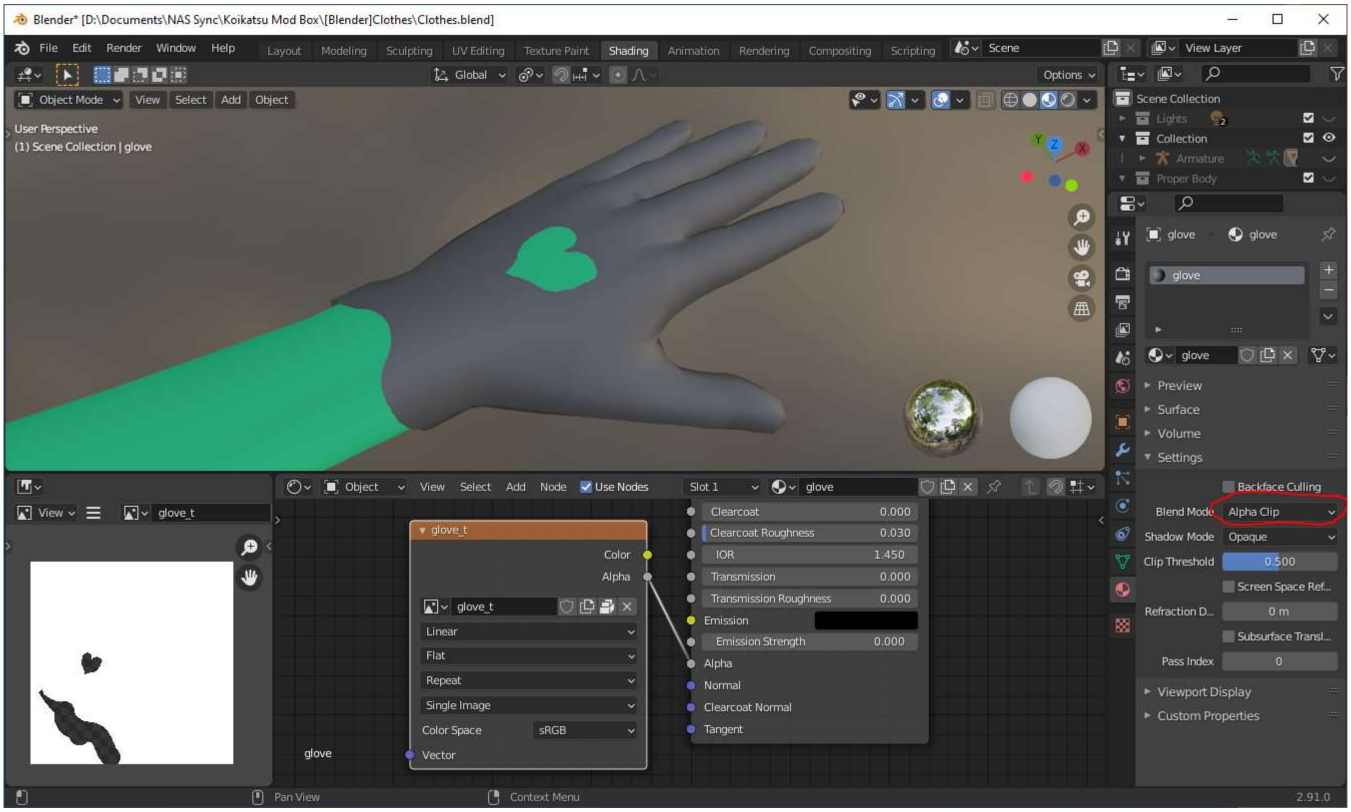
This is from some co-located vertices that get moved in different directions (the body mesh is split); you need to merge them together first (Mesh->Clean Up->Merge By Distance). The below picture is .002 with the Shrink/Fatten tool. You can go closer to the skin, but you'll get tiny amounts of clipping (maybe too small to notice in-game). You could also pull out problem areas a little extra.



Test for clipping in Blender by posing bones.



If the mesh edges aren't exactly where you want the edges of your item, make it bigger than you want and use transparency on the MainTex to determine where the edges of the item are (just like game bras). Texture paint transparency with Erase Alpha / Add Alpha brush setting. You'll have to set up the material in Blender to hide transparent parts of the mesh. Use Alpha Clip for it to work like the main_opaque shader, or Alpha Blend for it to work like the main_alpha shader.



Similar to Emblems, in Unity you may want to turn off shadow casting for skin-tight clothes so it is harder to notice they're floating above the skin, but then again you might want the shadow on the skin for the illusion of thickness. Normals and texture shadows also help.

_low Version

Clothes need a _low version, sort of like how Hair does, but it can be significantly more work because the _low character skeleton does not have all the same bones as the regular character skeleton; a separate skinning is required. Clothes also usually use a lower-poly mesh, unlike Hair. Don't worry about the _low version until you are done with the regular version. _low Clothes don't have half states.

There is a _low version for the opaque shader, but not the alpha shader or the emblem shaders. For alpha you still make a _low material to use lower resolution textures. For emblem you just use the same material.

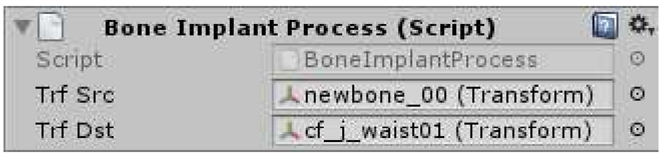
The game has some method for applying color picker colors to the _low version of the item; I don't know exactly how it works. Test your item in school (with non-default colors) to make sure it works properly. There's a thing where your item will be white if there's a problem, so don't use white for testing.

Adding Extra Bones

Clothes are normally limited to the body's bones, which means any dangling piece of clothing has to be weighted to skirt bones or something else. But the **ModBoneImplantor** plugin allows extra bones that you add to the skeleton to be copied to the body skeleton in-game (normally they aren't).

Things to do:

- in Blender add the extra bones and skin your item to them
- in Unity add the **BoneImplantProcess** MB to your item for each new bone chain
    - **trfSrc** is the first new bone in a chain
    - **trfDst** is its parent bone
- add DynamicBone MB for each new bone chain (or else they won't move)



Accessory Clothes

Accessory Clothes are Accessories with the **KK_AccessoryClothes** MB (and the Accessory MB, not the Clothes MB) which allows them to be skinned to the character's skeleton like Clothes are (requries the KK_AccessoryClothes plugin). This lets you make an Accessory that better moves with the body, but is not a good substitute for Clothes; as far as I know they get no Clothes features:

- no clothes color pickers; no patterns, need to use Accessory shaders for color pickers
- no features associated with the Clothes slots/shaders; no half states, AlphaMask, or liquids
- no features associated with the Clothes MB; no emblems / optional parts / built-in accessories

That's a lot of negatives, but here's some positives

- Accessories are not a mutually exclusive choice with other items like it would be if you put it in a Clothes slot
- you don't get half states, but you can make separate versions of the item so that your mod's users can show/hide with Clothes states via either the **KK_AccStateSync** or **Accessory States** plugin

From what I've seen, Accessory Clothes still use a Clothes hierarchy (no N_move) and **Armature Root** is filled in with **cf_j_root** like this on the MB:

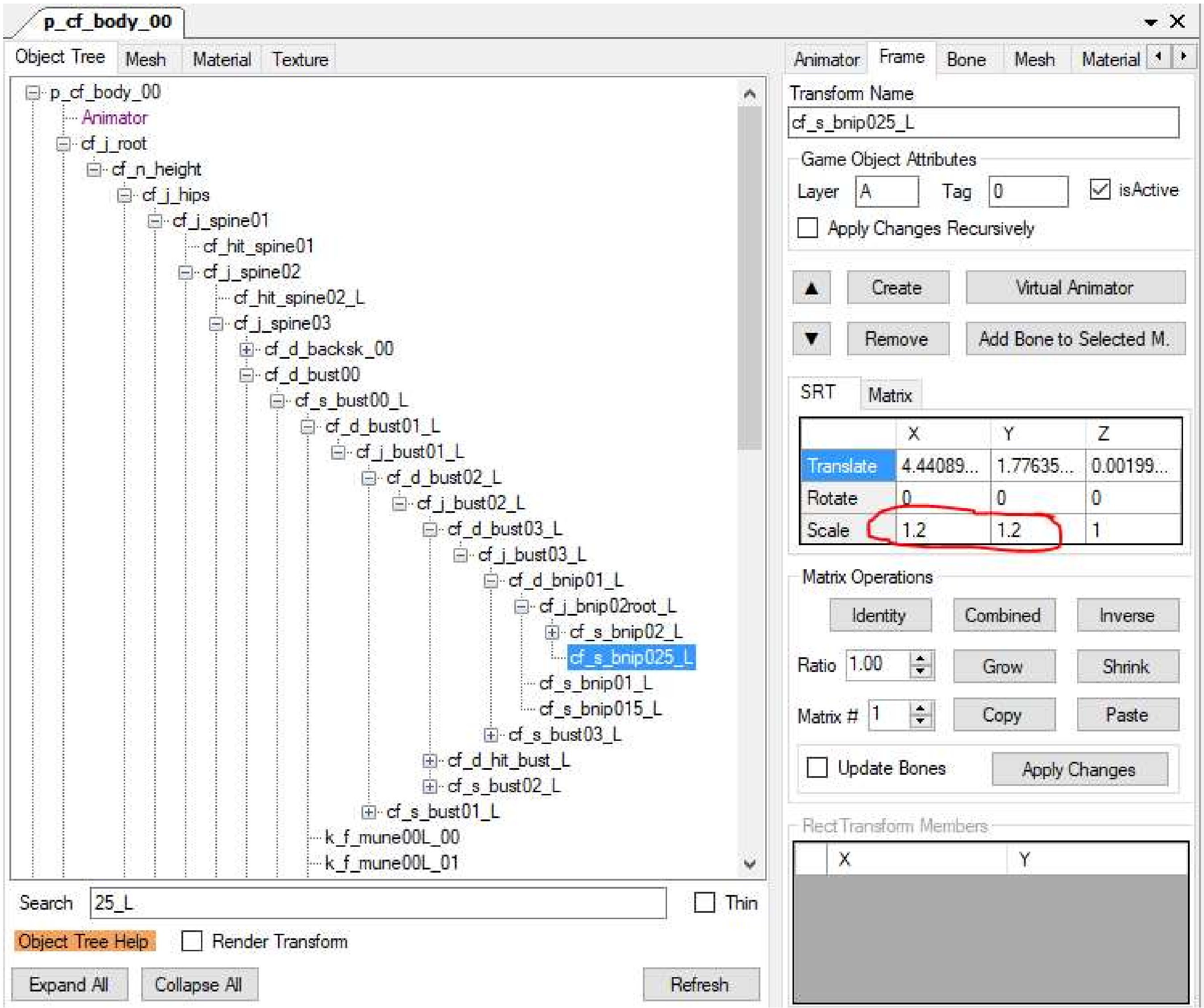
I don't know if it matters what accessory parent you specify in the list file, or the category (pick a category that makes sense for your item).

Preparing Blender: Clothes Version

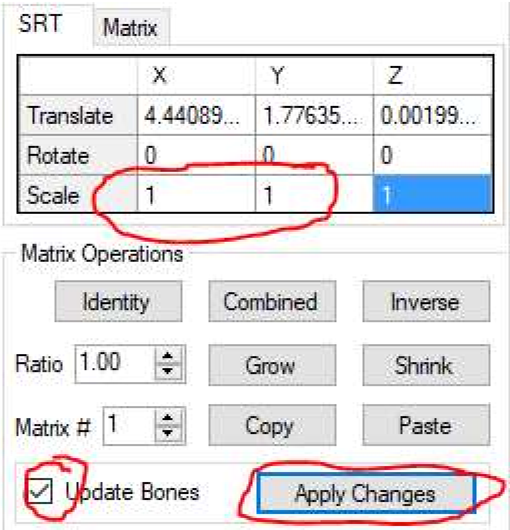
So obviously we will need the body in Blender (again) for making Clothes, but the focus this time will be on nipples because there is a special step to make them look right. When making tops and bras, you will often just AlphaMask away the skin underneath so that you don't have to worry about it clipping through. But even with the correct weight painting, your item's nipples can still end up with a weird shape depending on how you attach it to the body skeleton, and AlphaMask doesn't help you anyways if you want to make see-through Clothes.

So to make sure our process for Clothes is good, we will bring a game bra into Blender, delete its skeleton and attach it to the body skeleton, and turn that into a mod. We know the mesh is fine (in particular, the bone weights), but if our process to attach it to the body skeleton is bad, it won't look right in-game. Here's an example of how the nipples can be messed up:

The special step for getting nipples to look right is actually the first thing you do, before you export the body mesh with SB3U: you must change the scaling on a couple nipple bones which apparently Illusion changed outside of modeling software. So look at the body's hierarchy in SB3U and find the **cf_s_bnip025_L** bone. You will see 1.2 scaling on X and Y.



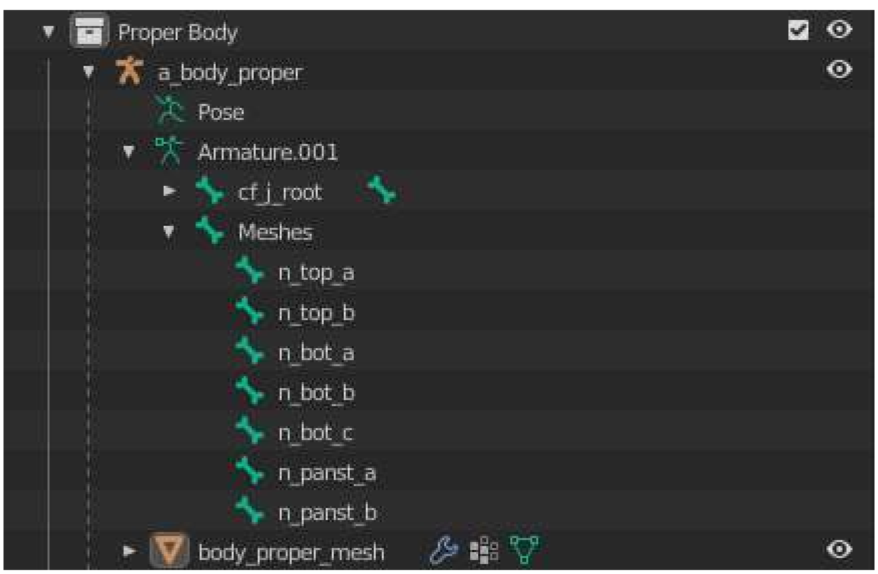
Change the scales to 1, check the Update Bones box, and Apply Changes.



Then do the same thing on the **cf_s_bnip025_R** bone.

Then you're good to export the body mesh with All Frames (which you need for skirt bones).

In Blender, set the armature scale to 1 as usual. Delete p_cf_body_00, then cf_o_root and all of its children. Start the hierarchy (which you'll also want to have in a bone layer for parenting meshes) with a Meshes bone (to stay consistent with my other guides). Make appropriately-named child bones for half states. I also did some renaming, for organization.



You'll want to put bones in different layers like I did in the Accessory guide, but this time the bones you want to have easy access to are:

- all cf_j bones (for posing to test for clipping)
- the Meshes bone and its children (for parenting meshes to)
- all bones you might skin clothes to (for weight painting; basically, the ones used by the body mesh plus the skirt bones; many of these are cf_j bones) but I don't know an easy way to select them all so you might want to just hide stuff like cf_pv and cf_hit and a_n instead

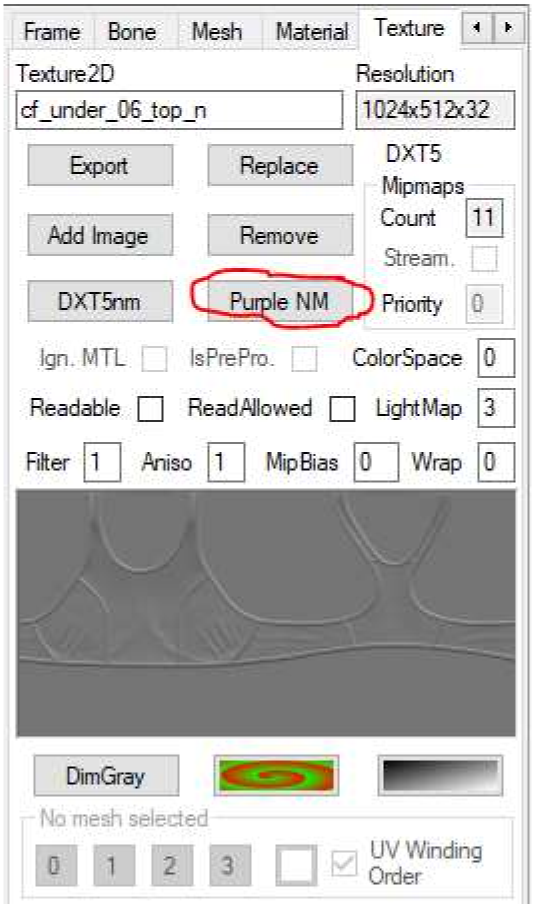
You could go through deleting bones that Clothes will never need, but you might make a mistake; it is safer to let a script delete them later.

Set up the body material however you want. I just made it green so it is easy to see clipping.

You will eventually need the _low body to make a _low version of your item. Everything is basically the same but there's no nipple bones to deal with. You can wait until you're happy with the high poly version though.

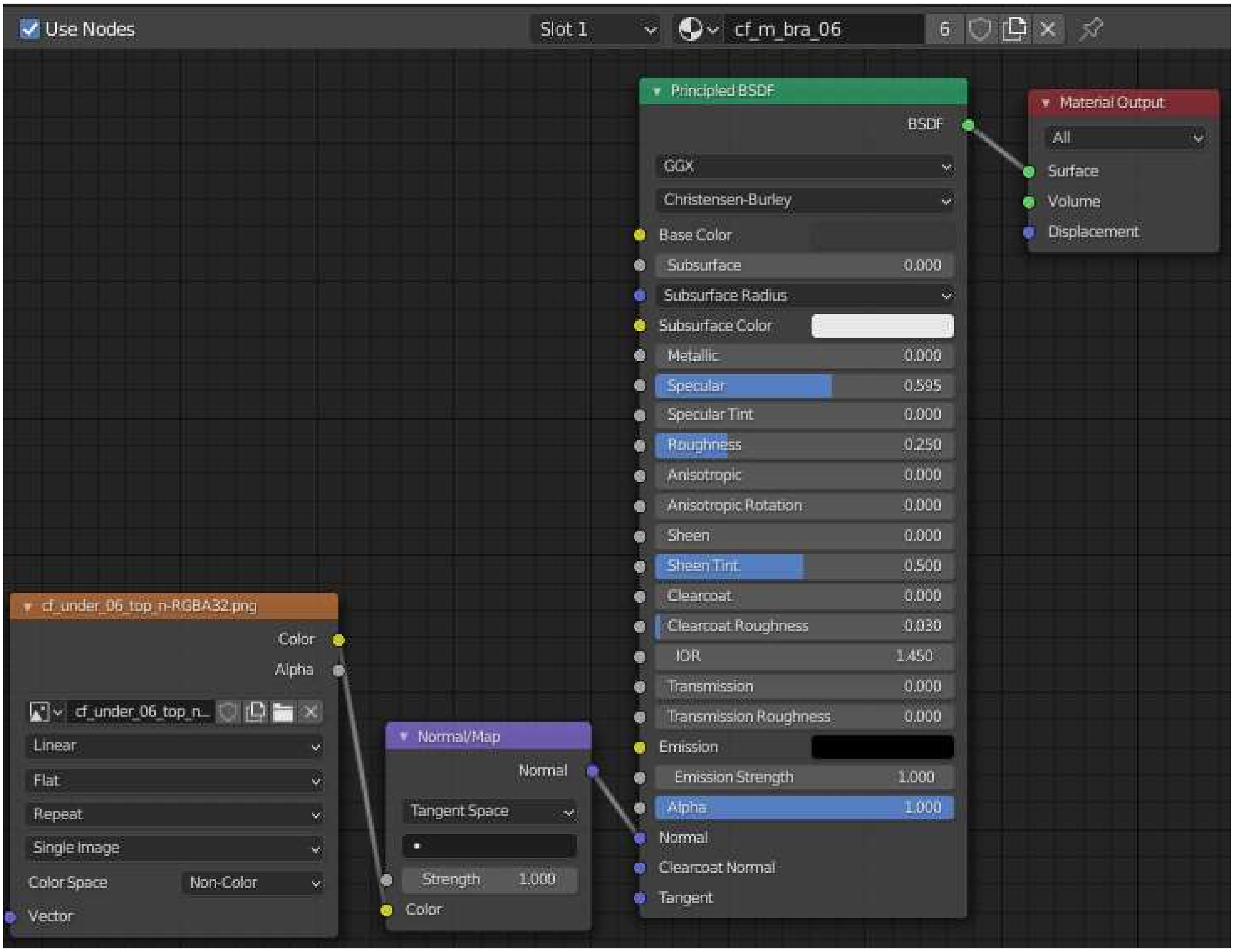
General Clothes Process: Blender

So now I'm ready to grab a game bra mesh for this test. I'll also grab the normals since I haven't dealt with normals in any of my 3D item guides so far. A few months ago SB3U gained the feature of being able to convert the "KK style" normal maps to the regular blue-ish format (otherwise it's a bit of a pain to convert). So click the Purple NM button and you'll see it convert.

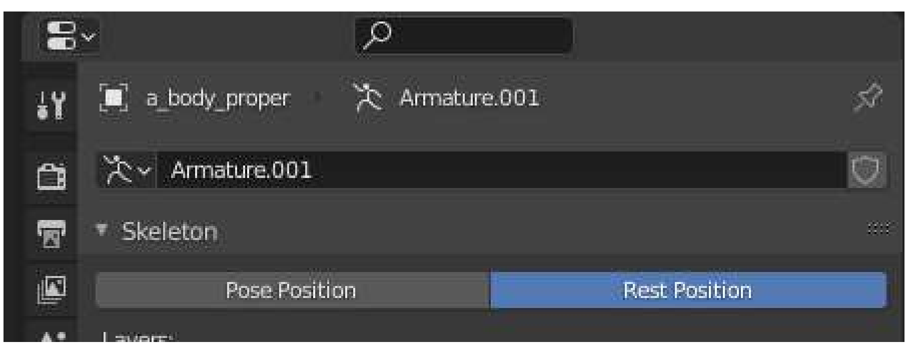


Exporting it will make a bmp which Blender doesn't seem to like, so you'll have to resave it as a png with GIMP. So now you can bring that bra mesh with blue-ish normals into Blender. Bring the bra's armature scale up to 1 so it is in the right place, and on the mesh do Alt+P (Clear Parent and Keep Transformation). Remove the Armature modifier and delete the bra's armature. The bra mesh should now be located perfectly on the body with no clipping; if you were porting a Clothes item you would have to do some adjusting first to put the mesh on the KK body, and if it required posing you would Apply the Armature modifier instead of removing it.

Set up the bra's material to use our blue-ish normal map. Since this bra is just to test nipples, I'm not going to do anything with other textures (and the normal map doesn't actually affect clipping either).

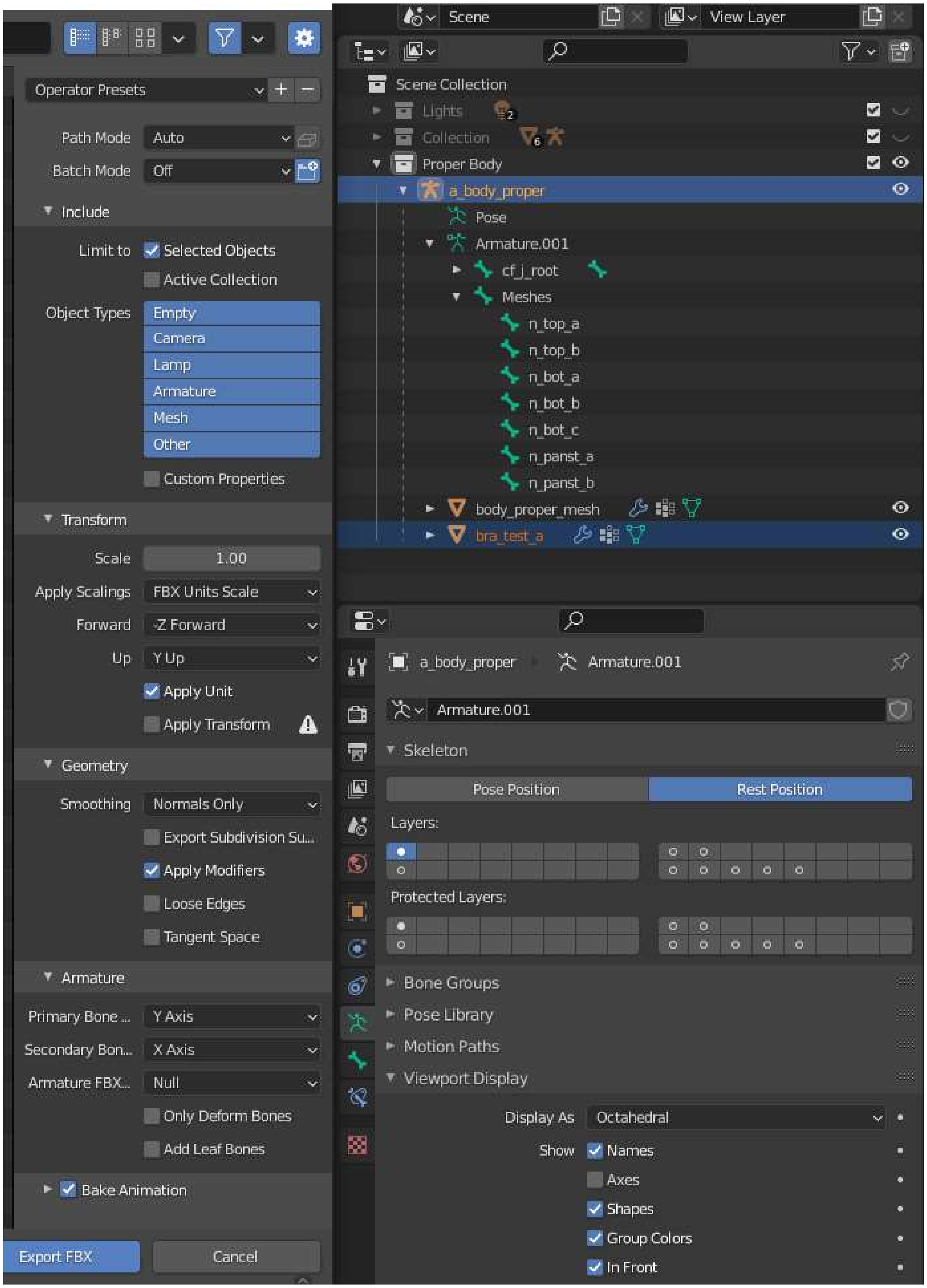


Make sure the body's armature is in rest position.



Then parent the bra mesh to n_top_a and put an Armature modifier back on it (now using the body's skeleton). You can set the armature back to pose position and pose to check for clipping, but there's a lot of small bones in the boobs so it's a pain. Rename the bra mesh to something that makes sense.

So when you're ready to export the mesh, in object mode set the armature back in Rest Position and use select the item's mesh(es) and the armature. I use the same export settings as in my other guides, except this time I am exporting Selected Objects instead of Active Collection, so that I don't get the body mesh.



Preparing KK Modding Tools: Clothes Version

KK Modding Tools has a Clothes example you can copy; do the usual manifest setup and delete all the example stuff.

KK Modding Tools isn't really setup to give proper clothes previews. It doesn't emulate how the game sends a color-mixed texture into MainTex, but you can manually put a texture in MainTex to check it out. It shows all states of an item at the same time, but you can manually go in and disable them. But these things may mess up your mod if you forget to revert them, so I don't even bother. Basically you want to be pretty confident you got everything right in Blender because your progress will be slow if you don't know there is a problem until you are testing the item in-game.

As I mentioned earlier in the guide, I couldn't get optional parts to work with the Clothes MB that comes with KK Modding Tools. If you want to edit the script to match the old Clothes MB, now is the time. I posted it in #mod-modelling on the KK Discord.

I made a script called **DeleteExtraBones** which deletes a bunch of extra stuff from your item. It is not necessary but helps make your mod a little smaller. A good place to put it would be in Assets/Scripts (where the MBs are). You can find this script also in #mod-modelling on the KK Discord. Explanation of exactly what it does and how to use it later.

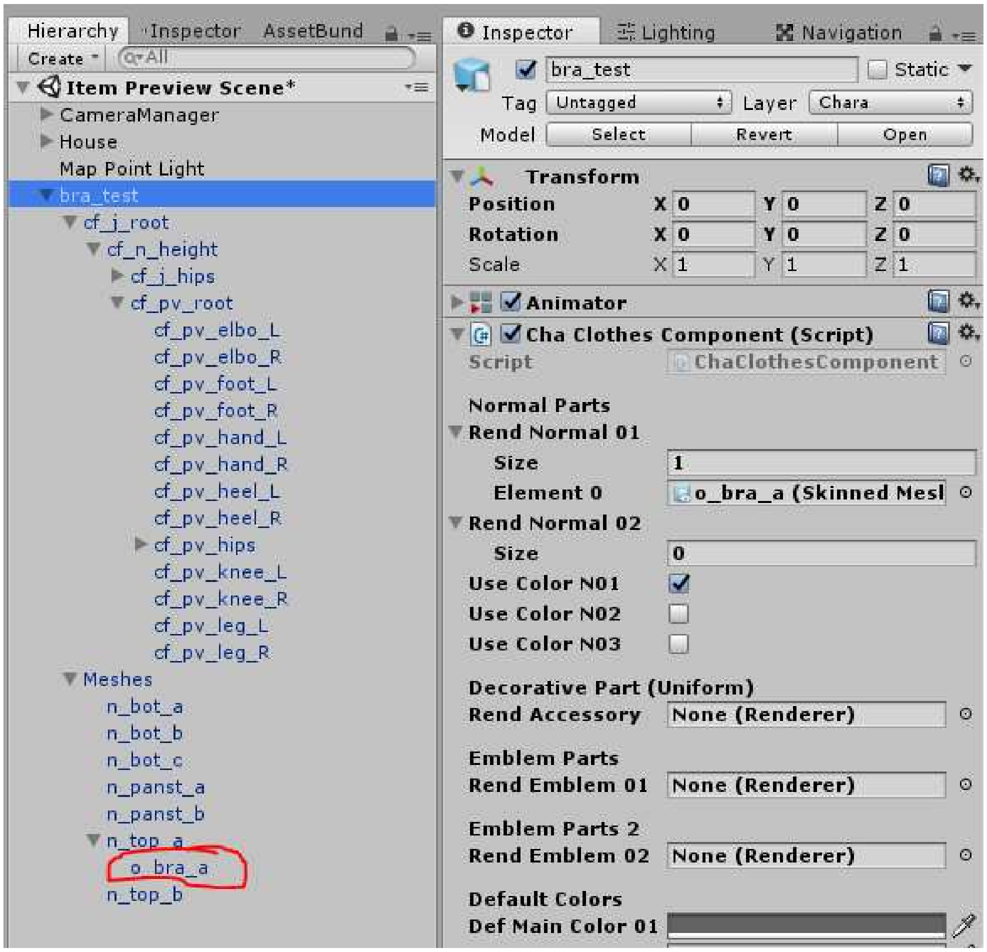
General Clothes Process: Unity

The Clothes process in Unity should make a lot of sense if you made Studio Items, Accessories, and Hair:

- Import textures and FBX
- make material and fill it out
- drag FBX into hierarchy
- drag material onto mesh
- put Clothes MB on item and fill it out (or for Accessory Clothes, that MB plus Accessory MB)
- add any other components such as DynamicBone and BoneImplantProcess
- (to be continued)

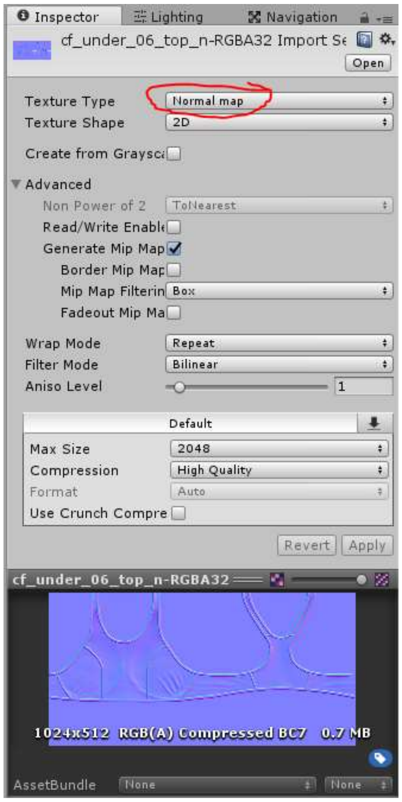
You may come across information that says to scale the cf_s_bnip025_L and cf_s_bnip025_R bones back to what they were before we changed them. I tried with and without doing this and couldn't find an in-game difference, so don't worry about it.

Because the body skeleton comes into Blender with 90 degree X rotation (and we didn't touch it), the Unity rotation zeros it out; we are happy with this. But what we will do in this case is rename the mesh's GameObject to o_bra_a so that it will receive the bra mask in-game (or else the nipples will poke through most tops). You could have named it that in Blender, but Blender doesn't let you have multiple meshes with the same name so you'd only be able to have one bra in the file.

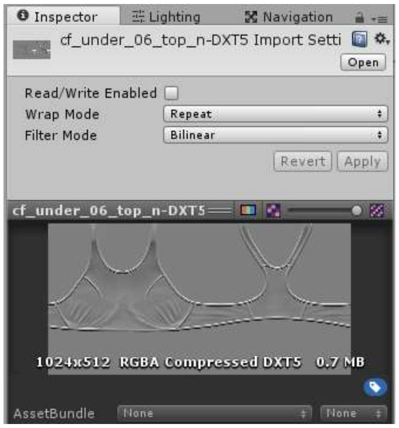


Again, I'm not going to use the original item's textures, just the normal map. So I'm just going to use a solid red color mask (and white main tex) so only 1 color picker.

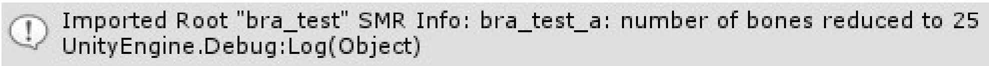
Normally when you have a blue-ish normal map, Unity will secretly convert it to the gray "KK style" normal map when you set the texture type to Normal map in the import settings. KK Modding Tools further changes this to the red style normal map for forward compatability with some other games (still works in KK). It will still look blue-ish in the preview, but don't worry, you can safely put it on shaders (including the KK shaders).



But in this case, we are compressing an already-compressed normal map, so it is better to just use the original gray normal map. You can just export the original (in gray format) with SB3U and import the dds into Unity. You don't get the same import settings for dds, so you can only do this when it is in the exact format you want.



You probably noticed our item has a whole bunch of extra GameObjects ("bones") it doesn't need. Thanks to a script in KK Modding Tools, when we import an FBX the skinned mesh renderers remove their references to bones with no bone weights.

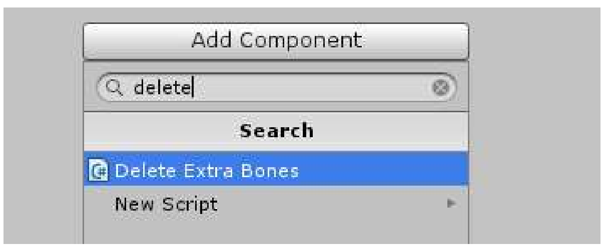


So now it's time to put the DeleteExtraBones script on your item. It immediately deletes all "deletable" GameObjects in the item, which is if the GameObject

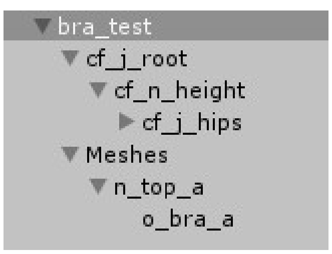
- does not have bone weights
- does not have any components besides Transform
- does not have any child GameObject that isn't deletable

Then it deletes itself from the item so you don't have to do it yourself.

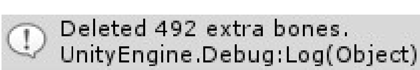
To use, just add the script to your item.



You won't even see it show up because of the self-deletion, but you will see a lot of bones / hierarchy stuff is now gone.



And you'll see it tell you in the console how many GameObjects were deleted.



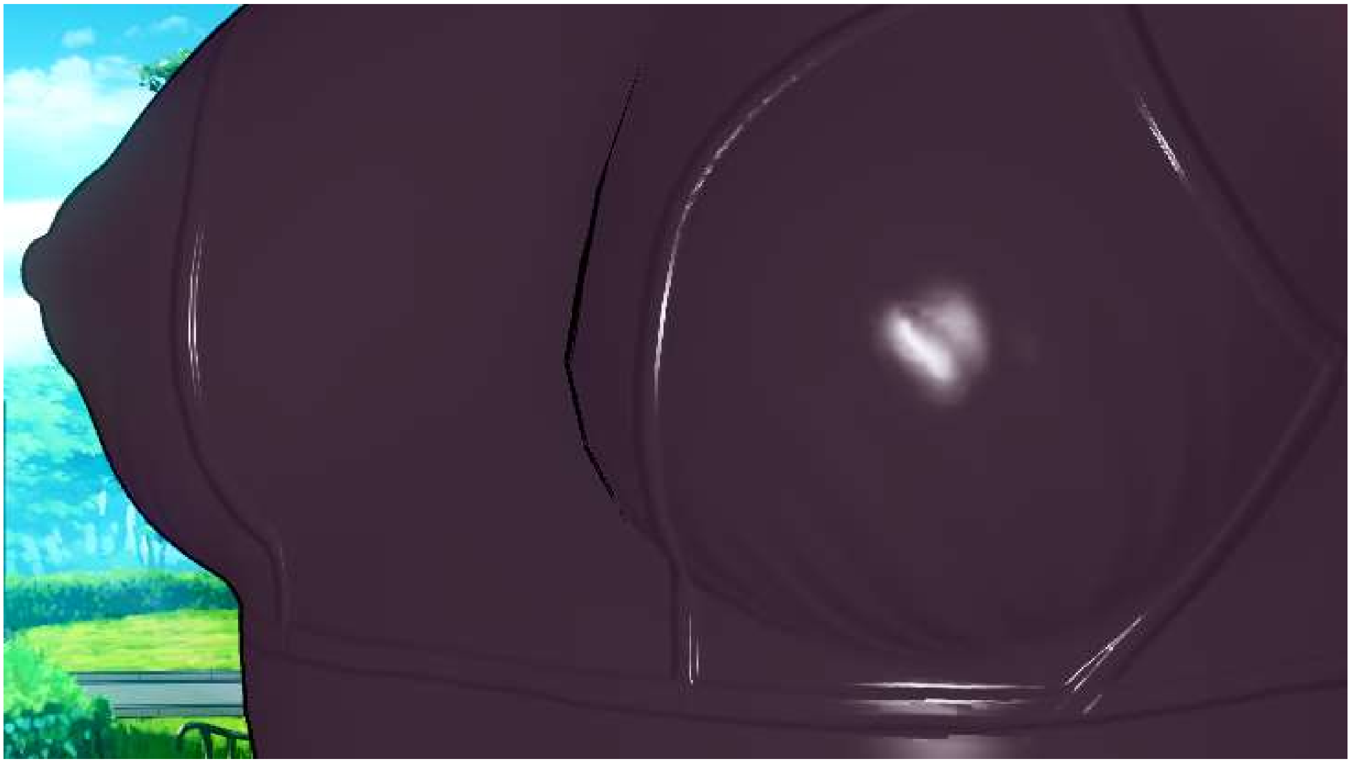
Anyways, to finish making the item:

- make the prefab and delete from Hierarchy window
- assign prefab and anything else needed (MainTex, ColorMask) to AB
- if you have _low mesh prepared, do it all again to make _low prefab
- list file entry (don't put any masks in the list entry yet; we want to see if there's clipping)

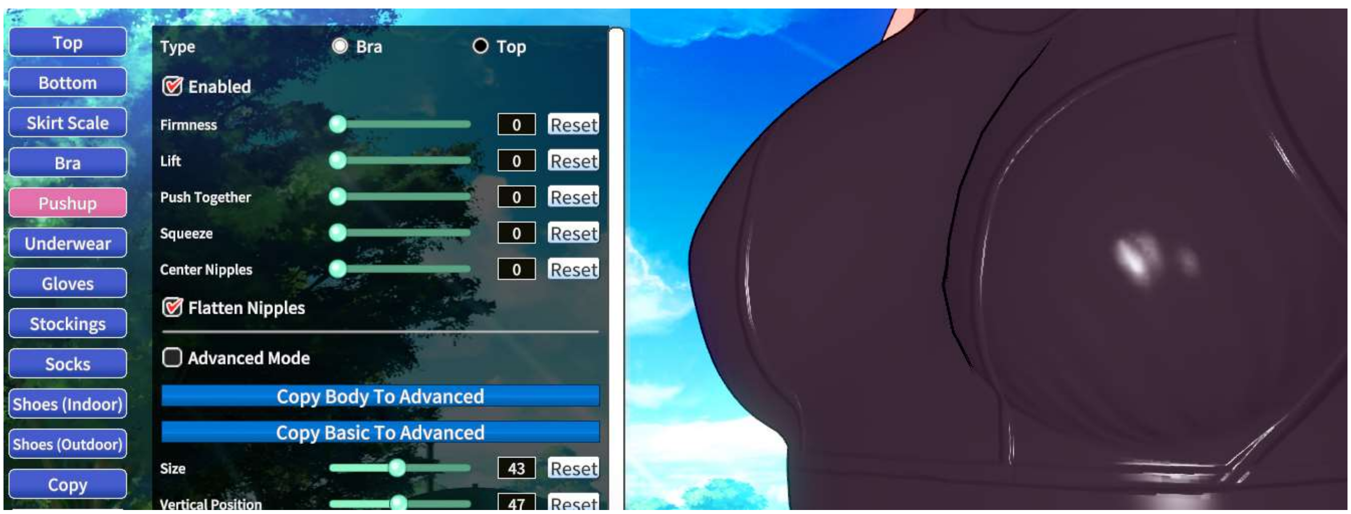
Build ABs, build Zipmod.

General Clothes Process: Result

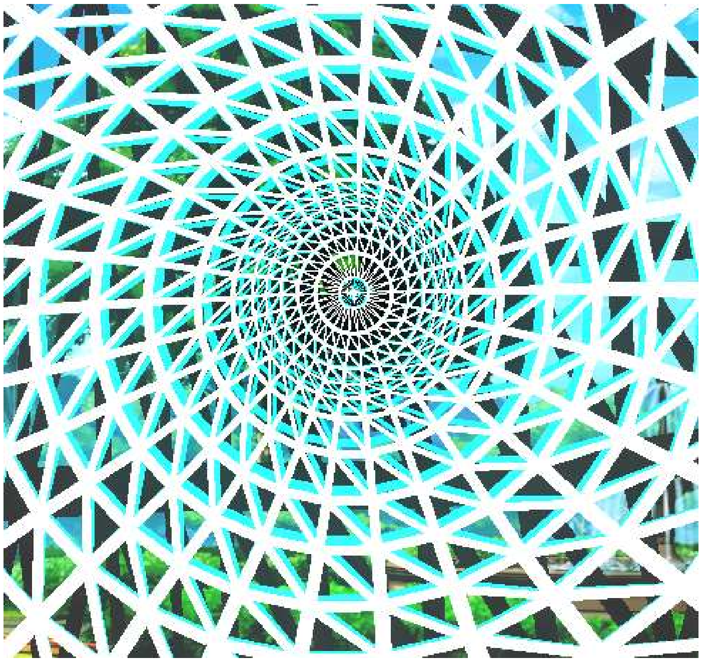
So now we've got a nice full-nipple bra without clipping.



Or if you don't want the full nipple, use the **Pushup** plugin to flatten the nipples (and whatever other boob shape changes you want). Otherwise to get flat nipples you have to remove nipple bone weights in Blender, and use a body mask to hide the nipples which would be poking through.



If you compare this (white wireframe) to the original bra (cyan wireframe), you'll see there is a slight difference around the nipples (like 0.1 mm), so technically the process is still imperfect. But it looks fine and I haven't found any clipping (that the game bra doesn't have), so I'm happy with it.



If your bra still has the half state (based on StateType in the list file) the mesh should disappear when you put it in half state because it is under n_top_a.

Also make sure your item is receiving the bra mask from the top (it should because we renamed the GameObject with the mesh to o_bra_a).